

DST-TransitNet: A Dynamic Spatio-Temporal Model for Robust Station-Level Transit Ridership Prediction

Jiahao Wang^{ID} and Amer Shalaby^{ID}

Abstract—Accurate short-term prediction of station-level transit ridership is crucial for effective transit operations, especially in dynamic urban environments. Traditional methods (e.g., ARIMA, SARIMA) fail to capture intricate spatial-temporal correlations and handle dynamic ridership fluctuations. This study introduces DST-TransitNet, a hybrid deep learning model integrating Graph Attention Networks (GAT) and Gated Recurrent Units (GRU) to model spatial-temporal dependencies among transit stations dynamically. A time-series decomposition framework further improves prediction accuracy and interpretability. Evaluated on Bogotá’s BRT system under normal operations, social disruptions, and pandemic conditions, DST-TransitNet achieved up to 94% R^2 and reduced prediction errors (MAAPE) by approximately 10%-15% compared to state-of-the-art benchmarks. These results demonstrate DST-TransitNet’s superior accuracy, robustness, and scalability across diverse prediction scenarios and changing time intervals.

Index Terms—Transit planning, real-time transit management, deep learning, transit ridership prediction, transit data analysis.

I. INTRODUCTION

ACCURATE prediction of transit ridership at fine spatial resolution is vital for efficient public transport planning and real-time operations. In rapidly growing urban areas, unexpected surges in passenger demand can lead to overcrowded vehicles, longer boarding times, and system-wide service disruptions. Timely and reliable short-term ridership predictions enable transit agencies to proactively allocate resources, adjust vehicle deployment, and optimize scheduling, thereby maintaining service quality and operational resilience.

Traditional time-series models such as Autoregressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA) have been widely applied to transit ridership prediction tasks. However, these classical models have notable limitations. Specifically, they struggle to accurately capture short-term fluctuations due to their inherent inability to model rapid temporal changes in ridership patterns dynamically. Moreover, they treat stations as independent entities, overlooking the critical spatial correlations among neighbouring

or functionally connected stations. This omission limits their ability to model the non-linear and evolving nature of urban transit systems [1].

To address these limitations, recent studies have turned to deep learning (DL) approaches [2]. DL models—particularly those leveraging convolutional neural networks (CNNs), recurrent neural networks (RNNs), and graph neural networks (GNNs)—have shown superior capabilities in capturing complex spatio-temporal relationships in transit networks. Nevertheless, three major challenges remain:

- **Dynamic Spatio-Temporal Feature Extraction:** Most existing DL models rely on static spatial structures (e.g., fixed adjacency matrices), which fail to capture evolving spatial dependencies throughout the day or under unexpected disruptions.
- **Balancing Accuracy and Computational Efficiency:** Many DL models are computationally intensive, limiting their feasibility for deployment in resource-constrained transit agencies.
- **Ensuring Scalability and Robustness:** Predictive models must remain accurate and stable when exposed to novel conditions—such as pandemics or protests—and across varying forecasting horizons.

To address these challenges, we propose DST-TransitNet (Dynamic Spatio-Temporal Transit Network), a hybrid deep learning model designed explicitly for short-term, network-wide station-level ridership forecasting. DST-TransitNet dynamically integrates spatial and temporal dependencies using Graph Attention Networks (GAT) and Gated Recurrent Units (GRU), supported by a time-series decomposition module that improves interpretability and predictive precision.

The key contributions of this paper are as follows:

- **Dynamic Spatial Correlation:** We propose a novel autonomous mechanism for interpreting dynamic spatial correlations using GAT. This enables the model to learn and adapt spatial dependencies among all stations from historical data via attention-based graph representations.
- **Temporal Feature Interpretability:** We incorporate a time-series decomposition-based temporal embedding layer to improve interpretability. This enhances transparency by clarifying how temporal factors influence model outputs.

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The authors are with the Department of Civil and Mineral Engineering (Transportation Engineering), University of Toronto, Toronto, ON M5S 1A4, Canada (e-mail: jhope.wang@mail.utoronto.ca).

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TABLE I
STATISTICAL MODELS FOR TRANSIT RIDERSHIP PREDICTION

Citation	Key Factors Considered	Horizon	Real-Time
Cervero et al. (2010) [5]	Built environment, service frequency	Yearly	No
Gutiérrez et al. (2011) [6]	Land use, accessibility, distance decay	Monthly	No
Diab et al. (2020) [7]	Revenue km, car ownership, policy, service factors	Yearly	No
Liu et al. (2016) [8]	Station environment, service frequency	Daily	No
Stover & McCormack (2012) [4]	Weather (wind, temp, rain, snow), route frequency	Daily	Yes
Deepa et al. (2022) [3]	Frequency, inter-route dynamics, catchment overlap	Peak (short-term)	No

- **Spatio-Temporal Fusion Architecture:** We present a hybrid deep learning architecture that captures dynamically evolving spatial relationships and temporal patterns through an integrated GAT-GRU structure.
- **Robustness Across Operational Scenarios:** We conduct extensive evaluations using real-world data from Bogotá's Bus Rapid Transit (BRT) system,¹ covering normal operations, protest events, and COVID-19 disruptions. DST-TransitNet consistently outperforms benchmark models in both accuracy and robustness.
- **Scalability Across Prediction Intervals:** The model demonstrates strong and stable performance across different prediction horizons, outperforming baselines in both short- and long-term settings.
- **Real-World Deployment Insights:** We outline a deployment framework to facilitate practical adoption of DST-TransitNet, highlighting its feasibility for real-time, station-level forecasting in operational environments.

The remainder of this paper is structured as follows: Section II provides a concise literature review, highlighting limitations of existing ridership prediction methods. Section III presents the detailed methodology of DST-TransitNet. Section IV describes the experimental setup, datasets, and analysis of results under diverse conditions. Finally, Section V summarizes key findings and outlines future research directions.

II. BACKGROUND

This section reviews state-of-the-art public transit ridership analysis techniques and general time-series forecasting. It provides an in-depth overview of key methodologies, applications, and limitations, focusing on challenges relevant to this study.

A. Statistical Methods for Transit Ridership Analysis

Statistical modelling is widely used for ridership estimation, primarily through direct ridership models and travel demand models [3]. These methods assess how socio-economic factors, land-use characteristics, built environment variables, and transit service attributes influence transit demand. Table I summarizes several representative studies, most focusing on longer prediction horizons (e.g., yearly, monthly, daily) suitable for strategic planning or policy evaluation—particularly for proposed or evolving transit systems.

A key limitation of statistical models is their strong reliance on carefully selected exogenous variables, which limits their transferability across different regions or transit systems. To improve flexibility, some researchers have incorporated real-time variables (e.g., weather conditions) [4] and explored spatial dependencies such as inter-route interactions [3].

TABLE II
DEEP LEARNING MODELS FOR TRANSIT RIDERSHIP PREDICTION

Citation	System / Region	Level	Model	Horizon	Key Insight
Liu et al. (2019) [11]	Nanjing Metro	StL/10min	LSTM + FFNN	10 min	Hybrid model enhances short-term metro accuracy
Cheng et al. (2022) [10]	Kaohsiung Bus	RtL/daily	FFNN, SVR, RBF	Next day	LSTM outperforms ARIMA for daily bus prediction
Bai et al. (2019) [12]	Shenyang Transit	RL/60min	CNN + LSTM	1 hour	Joint learning of spatial-temporal OD patterns
Li et al. (2020) [13]	Sydney Transit	RL/15min	PGCM (GCN)	15–60 min	Graph adjacency improves OD prediction accuracy
Wang et al. (2022) [15]	Beijing Subway	StL/5min	GCN-Au-GRU	5–15 min	Time-varying spatial graphs improve performance
Wu et al. (2023) [14]	Nanning Metro Line 1	StL/15min	MFGCN + ResNet	15–60 min	Multi-task prediction boosts overall accuracy
Da Silva & Shalaby (2024) [16]	Toronto Subway Line 1	StL/1hr	LR, ARIMA, RF, T-GCN	1–18 hr	Wi-Fi-based study comparing classical and DL models

Note: StL = Station-Level, RL = Region-Level, RtL = Route-Level

Nonetheless, these models are still constrained by limited adaptability and infrequent updates, which restrict their usefulness for dynamic, short-term applications.

B. Machine Learning and Deep Learning for Transit Ridership Prediction

To overcome the rigidity of statistical models, researchers have increasingly adopted machine learning (ML) and DL methods, which offer greater flexibility for modelling short-term dynamics and capturing complex spatio-temporal patterns.

ML/DL models can be trained directly on historical ridership data, which transit agencies routinely collect. This allows for rapid updates and reduced dependency on manual data collection [9]. Table II highlights recent studies utilizing various neural network architectures.

Feedforward neural networks (FFNN) and support vector regression (SVR) [10] are commonly used to model temporal trends. Recurrent models like LSTM and GRU improve the learning of sequential dependencies [11]. To capture spatial features, CNNs [12] and graph convolutional networks (GCNs) [13] have been employed—the latter being particularly effective in handling the non-Euclidean structure of transit networks. Attention mechanisms have also shown promise in enhancing temporal feature extraction and interpretability [14].

Compared to statistical models, ML/DL approaches enable high-resolution prediction and simultaneous forecasting for multiple stations. However, several challenges remain: (1) shorter time intervals increase noise and reduce prediction accuracy [13]; (2) generalization varies across stations due to differing local dynamics; and (3) scalability becomes a concern as forecast horizons expand [11]. Furthermore, GCN models relying on fixed adjacency matrices may struggle under dynamic transit conditions [16], where ridership flows vary significantly throughout the day.

C. Advanced Time-Series Forecasting Models

Beyond transit-specific studies, the broader field of time-series forecasting has witnessed significant progress. Table III presents several advanced DL models applicable to spatio-temporal prediction tasks.

Transformer-based models such as *iTransformer* [17] employ multivariate attention to flexibly model temporal patterns, while *DLinear* [18] improves efficiency by decomposing time series into long- and short-term components. *DT-GRU*

¹<https://datosabiertos-transmilenio.hub.arcgis.com>

TABLE III
COMPARISON OF SPATIO-TEMPORAL MODELS FOR
TRANSIT OR TRAFFIC PREDICTION

Model	Spatial	Temporal	Level	Graph	DC
iTransformer [17]	—	Multi-Attn	SL/5min	—	—
DLinear [18]	—	Decomp.	SL/5min	—	—
DT-GRU [19]	Transf.+GRU	—	SL/15min	—	—
GraphWaveNet [23]	DiffConv	DCCConv	SL/15-60	—	—
STD-Net [30]	CNN	LSTM	RL/30-60	Img2D	—
DHSTNet [20]	CNN	ConvLSTM	RL/30-60	Img2D	—
AAtt-DHSTNet [21], [26]	ST-Attn	ConvLSTM	RL/30-60	2D+VG	Attn
GCN-DHSTNet [22]	GCN	LSTM	RL/30-60	PC	GCN
GRUGCN [29]	GCN	GRU	SL/30-60	PC	—
STFGNN [27]	CNN	CNN	SL/5min	VG	GatedCNN
STFGCN [25]	GCN	LSTM	SL/5min	PC	—
STGNN [28]	DiffConv	GRU	SL/30-60	VG	NodeEmb
DST-TransitNet (Ours)	GCN+GAT	Decomp.+GRU	SL/15-180	PC+VG	GAT

Notes: SL = Sensor-Level, RL = Region-Level, StL = Station-Level, PC = Physical Connection, VG = Virtual Graph, DC = Dynamic Connectivity, Decomp. = Decomposition, Attn = Attention, Img2D = 2D Image

[19] combines dual transformer blocks with GRU units to unify spatial and temporal attention.

Graph-based modelling has also evolved significantly. The DHSTNet family [20], [21], [22] represents a transition from CNN-based to GCN-based spatial modelling, marking a shift toward graph-centric architectures. *GraphWaveNet* [23] applies diffusion convolution [24], while *STFGCN* [25] leverages fixed physical topologies. *AAtt-DHSTNet* [26] enriches regional graphs with virtual links and gated CNNs. More recently, *STFGNN* [27] introduces dynamically updated virtual graphs to reflect flow-based connectivity. *STGNN* [28] integrates node embeddings and diffusion convolution to represent global-local spatial features more efficiently. Gao and Ribeiro [29] further explore the expressiveness of spatio-temporal graphs, concluding that time-then-graph encoding provides superior representation for hybrid spatio-temporal feature extraction.

Our proposed model, DST-TransitNet, unifies physical connectivity and latent demand similarity within a hybrid GCN framework. Temporal dependencies are modelled using multivariate attention and GRU layers. A graph-attention mechanism dynamically updates edge weights in response to real-time ridership fluctuations, ensuring robustness across stations and under varying operating conditions (e.g., special events, pandemic disruptions), while remaining computationally efficient and scalable across multiple prediction horizons.

III. METHODOLOGY

This section presents the architecture of the proposed Dynamic Spatio-Temporal TransitNet (DST-TransitNet), a deep learning framework designed to overcome the limitations of existing ridership prediction models. DST-TransitNet integrates Graph Convolutional Networks (GCN), Graph Attention Networks (GAT), Gated Recurrent Units (GRU), and Feed-Forward Neural Networks (FFNN) to model complex spatial-temporal dependencies in transit demand data effectively.

A. Basic Building Blocks

We introduce the core components that form the foundation of DST-TransitNet: GRU, GCN, and GAT. These modules

are critical to understanding the model's architecture and functionality.

1) *Gated Recurrent Unit (GRU)*: The GRU, proposed by [31], is an enhancement over traditional recurrent neural networks (RNNs), designed to address the vanishing and exploding gradient issues. It updates the hidden state h_t using the previous state h_{t-1} and current input X_t , enabling it to capture temporal dependencies over time:

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t,$$

where the candidate activation $\tilde{\mathbf{h}}_t$ is:

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h \mathbf{x}_t + \mathbf{r}_t \odot (\mathbf{U}_h \mathbf{h}_{t-1}) + \mathbf{b}_h),$$

and the update and reset gates are:

$$\mathbf{z}_t = \sigma(\mathbf{W}_z \mathbf{x}_t + \mathbf{U}_z \mathbf{h}_{t-1} + \mathbf{b}_z),$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r \mathbf{x}_t + \mathbf{U}_r \mathbf{h}_{t-1} + \mathbf{b}_r).$$

2) *Graph Convolutional Networks (GCN)*: GCNs, introduced in [32], extend convolution operations to graph-structured data, allowing nodes to aggregate features from their neighbours:

$$H^{(l+1)} = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)} \right),$$

where:

- $H^{(l)}$ is the feature matrix at layer l ,
- $W^{(l)}$ is the trainable weight matrix,
- $\tilde{A} = A + I$ includes self-loops,
- $\tilde{D}$ is the corresponding degree matrix,
- σ is the activation function.

To enhance this, [33] proposed the k-dimensional GNN (k-GNN), which extends message passing to include non-adjacent nodes:

$$H^{(l+1)}(v) = \sigma \left(H^{(l)}(v) W_1^{(l+1)} + \sum_{w \in A} H^{(l)}(w) W_2^{(l+1)} \right).$$

3) *Graph Attention Network (GAT)*: GATs, proposed by [34], incorporate attention mechanisms into GCNs, allowing the model to learn the importance of each neighbour dynamically. The attention coefficient is computed as:

$$e_{ij}^{(l)} = \text{LeakyReLU} \left(\mathbf{a}^{(l)T} \left[\mathbf{z}_i^{(l)} \parallel \mathbf{z}_j^{(l)} \right] \right),$$

where $\mathbf{z}_i^{(l)} = \mathbf{W}^{(l)} \mathbf{h}_i^{(l)}$, and the final edge weights are:

$$W_{E_{ij}}^{(l)} = \frac{\exp(e_{ij}^{(l)})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik}^{(l)})}.$$

The updated node feature is:

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i)} W_{E_{ij}}^{(l)} \mathbf{z}_j^{(l)} \right).$$

B. DST-TransitNet Architecture

As illustrated in Fig. 1, DST-TransitNet consists of four key components: temporal decomposition, dynamic spatial weight calculation, spatio-temporal aggregation, and prediction. It leverages data from all stations, including current and historical passenger counts and spatial relationships, to produce system-wide forecasts.

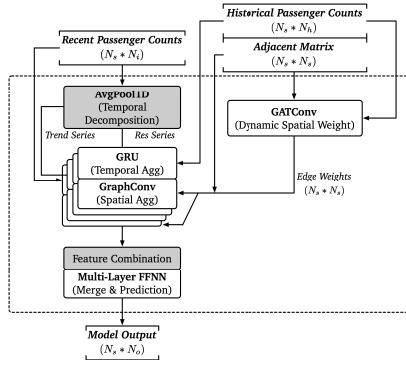


Fig. 1. Architecture of DST-TransitNet, which integrates temporal decomposition, dynamic spatial weighting, spatio-temporal aggregation, and a prediction module to generate station-level forecasts using recent and historical passenger data.

1) *Temporal Decomposition*: This layer decomposes the input time series X_o into trend X_t and residual X_r components using average pooling:

$$X_t = \text{AvgPool1D}(X_o),$$

$$X_r = X_o - X_t.$$

This improves feature interpretability and model performance.

2) *Dynamic Spatial Weight Calculation*: GAT is used to compute dynamic edge weights $\mathbf{W}_E$ based on historical ridership $\mathbf{X}_h$ collected at the same time in previous weeks. The input $\mathbf{X}_h = \{\mathbf{X}_{h1}, \dots, \mathbf{X}_{hN_h}\}$ consists of N_s stops with N_h records each. GAT computes:

$$(e, \mathbf{W}_E) = \text{GATConv}(\mathbf{X}_h, \mathbf{a}, \mathbf{w}),$$

where $\mathbf{a}$ is the adjacency matrix and $\mathbf{w}$ is the trainable parameter matrix.

3) *Spatio-Temporal Aggregation*: This layer combines stacked GRUs (for temporal dependencies) and k-GNNs (for spatial dependencies) to process X_o , X_h , X_t , and X_r . As shown in Fig. 2, each series is processed by a GRU followed by a graph convolution:

$$h_i = \text{GRU}(\text{series_agg}_i(\text{temp_inputs}_i)),$$

$$h_i = \text{k-GNN}(h_i, \mathbf{a}, \mathbf{W}_{E,ji})$$

$$= \mathbf{W}_1 \mathbf{x}_i + \mathbf{W}_2 \sum_{j \in \mathcal{N}(i)} \mathbf{W}_{E,ji} \cdot \mathbf{x}_j.$$

The final predictions are generated using a multi-layer FFNN that maps the fused spatial-temporal features to the output space.

4) *Long-Term Forecasting via Iterative Short-Term Prediction*: We extend our short-term (5-minute interval) prediction model for long-term (1)-hour forecasting using an iterative strategy as shown in Fig. 3. A pre-trained model generates successive predictions, updating its input window with each iteration:

- 1) **Initial Inputs**: Historical data $T_1, \dots, T_L$.
- 2) **Iteration 1**: Predict P_{I+1} from the initial window.
- 3) **Subsequent Iterations**: Shift the input window forward by adding the latest prediction and removing the oldest input.
- 4) **Output**: Collect predictions P_{I+1} to P_{I+N} for the full 1-hour forecast.

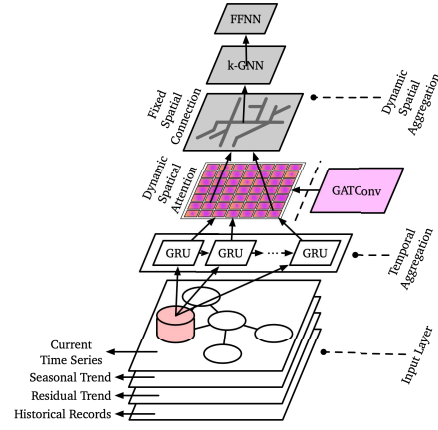


Fig. 2. The architecture of the Spatio-Temporal Aggregation layer in DST-TransitNet, combining Dynamic Spatial Aggregation (via k-GNN and GATConv) and Temporal Aggregation (via stacked GRUs).

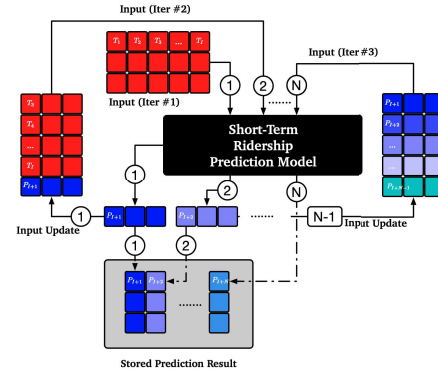


Fig. 3. Iterative strategy for long-term forecasting using a short-term prediction model. At each step, the model updates its input by incorporating the most recent prediction, enabling multi-step forecasts without retraining.

This method is resource-efficient and avoids model retraining. However, it introduces the risk of error accumulation, as each prediction relies partly on previous outputs. Ensuring robustness and stability is essential to maintain forecast accuracy.

C. Pipeline for Real-World Deployment

Fig. 4 illustrates the end-to-end pipeline for deploying DST-TransitNet in a real-world environment, comprising three phases: data preparation, model preparation, and application.

Data Preparation Phase:

- **Data Collection**: Collect recent and historical ridership data and station-level geographic information (e.g., latitude, longitude, and route attributes). While ridership data is updated dynamically, geographic data is relatively static and updated periodically.
- **Data Cleaning and Aggregation**: Preprocess the data by removing duplicates and aggregating ridership records within specified time intervals. Construct the adjacency matrix from station geographic data to represent the transit network structure.

Model Preparation Phase:

- **Data Splitting**: Divide the cleaned ridership data into training and testing sets.

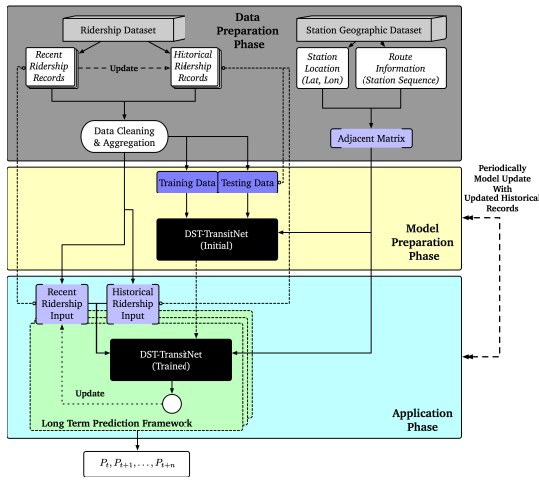


Fig. 4. End-to-end pipeline for real-world deployment of DST-TransitNet, comprising data preparation, model training, and application phases.

- **Model Training and Validation:** Train the DST-TransitNet model on the training data and validate its performance on the testing set.

Application Phase:

- **Input Data Requirements:** For each prediction at time $t + 1$, the model requires recent ridership data from $t - n$ to t and historical data from the previous day covering $t - n + 1$ to $t + 1$, where n is the model's input window length.
- **Long-Term Forecasting:** In the long-term forecasting mode, the model updates the recent input window using its own predicted output while retrieving historical data aligned with the future prediction time (e.g., $t + 2$, $t + 3$). Thus, only one current input is needed to forecast multiple future steps.
- **Continuous Learning:** Periodically fine-tune the model with newly collected data to ensure it adapts to recent trends and system changes, maintaining accuracy without full retraining.

This structured pipeline supports the efficient deployment of DST-TransitNet, enabling accurate, responsive, and scalable ridership forecasting for public transit systems.

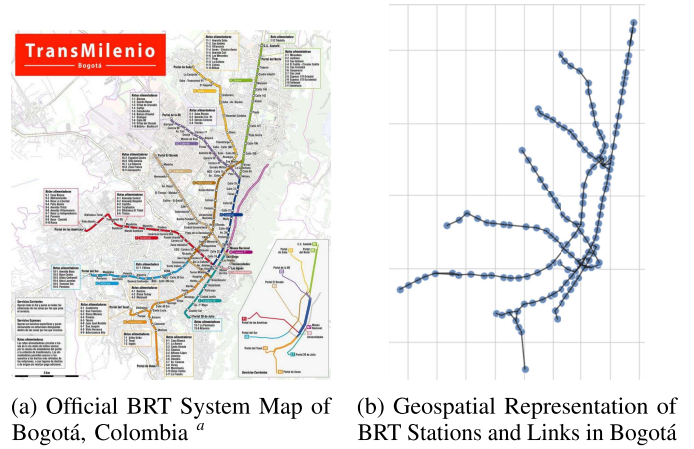
IV. EXPERIMENTS AND ANALYSIS OF RESULTS

This section presents the experimental framework used to evaluate the performance of DST-TransitNet. Specifically, we: (i) describe the datasets, (ii) define the baseline models and hyperparameter configurations, (iii) outline the computational environment and implementation tools, (iv) specify the evaluation metrics, and (v) discuss the empirical results.

A. Datasets

We utilize five years of Automatic Passenger Count (APC) data from the *TransMilenio* BRT system in Bogotá, Colombia, spanning from August 2015 to May 2021. Each record captures the **number of boardings** registered by fare collection equipment at 15-minute intervals for each station.² The network has 147 stations (nodes) and 320 directed links (edges)

²Data available at: <https://datosabiertos-transmilenio.hub.arcgis.com>



(a) Official BRT System Map of Bogotá, Colombia^a

(b) Geospatial Representation of BRT Stations and Links in Bogotá

^a<https://development.asia/case-study/efficient-city-transport-those-who-do-not-own-cars>

Fig. 5. Comparison of BRT System Maps: Official vs. Geospatial Representation.

across 12 BRT corridors. Before the COVID-19 pandemic, the system served an average of 2.3 million passengers daily, with over 250,000 passengers per hour during peak periods [35]. Figure 5 contrasts the official corridor map with a geospatial graph representation of the station layout.

Temporal Regimes. Following the classification in [9], we distinguish three demand regimes: *normal* (Aug. 2015–Nov. 2019), *protest* (Nov.–Dec. 2019), and *COVID-19* (Mar. 2020–May 2021). Figure 6 shows a sharp ridership decline after March 2020 attributable to the pandemic, as well as pronounced fluctuations in late 2019 caused by a transit-worker strike.

Train-Test Split. Consistent with [9], all data up to July 2018 (within the normal regime) are used for training. The remaining data serve as the test set, yielding three mutually exclusive evaluation subsets: normal, protest, and COVID-19.

15-Minute Forecasting Horizon. Unlike previous studies focusing on daily aggregates, we adopt a 15-minute forecasting horizon to facilitate real-time operational decision-making. Each input sample includes: (i) the 20 most recent 15-minute observations (i.e., a 5-hour look-back window), and (ii) 20 historical observations from the same time one week prior. This input design mirrors the requirements of DST-TransitNet, with the prediction target being the ridership value 15 minutes ahead.

B. Baselines and Hyperparameter Settings

DST-TransitNet (DS2TN) is compared against two conventional neural networks (FFNN and LSTM) and several recent state-of-the-art time series models:

- **iTransformer** [17]: An inverted-attention Transformer optimized for time-series forecasting.
- **DLinear** [18]: A decomposition-based linear model that shares temporal trend modelling similarities with DST-TransitNet.
- **GraphWaveNet** [23]: A spatio-temporal model combining dilated temporal convolutions and diffusion convolu-

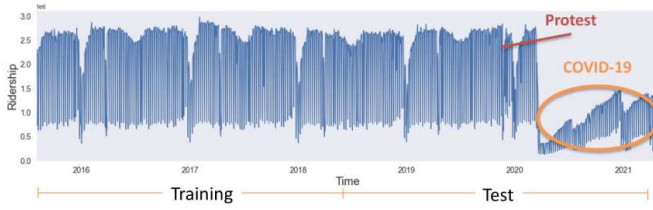


Fig. 6. Daily Aggregated BRT Ridership from August 2015 to May 2021. Training period: Aug. 2015 – Jul. 2018; Testing period: Aug. 2018 – May 2021. Protest: Nov.–Dec. 2019. COVID-19: Mar. 2020 – May 2021. (Adapted from [9]; Y-axis unit is 10^6).

tion GNNs. It stacks multiple residual blocks with skip connections and includes a trainable adjacency matrix.

- **GRUGCN [29]**: A sequential architecture that processes node-level time series using a GRU encoder followed by a GCN decoder. Skip connections and an MLP are used for final prediction.
- **STGNN [28]**: A time-first architecture using a shared GRU across nodes to capture temporal features, followed by a DiffConv GNN for spatial modelling. Trainable node embeddings enable both global and local representation.

To assess the contribution of individual modules in DST-TransitNet, we introduce two ablated variants:

- **DS2TN (w/o TD)**: DST-TransitNet without the time decomposition module.
- **DS2TN (w/ RDE)**: A variant where dynamic edge weights are based on recent observations instead of historical records.

We also propose **DST-TransitNet v.2** (DS2TN v.2), which relocates the GRU layer to the output prediction stage to improve computational efficiency.

Except for FFNN, LSTM, and iTransformer—which operate at the station level—all baseline models are trained on the complete set of 147 stations jointly. Key hyperparameters are summarized in Table V. All models are tuned via grid search on a validation set to ensure fair comparison.

C. Computing Environment

All experiments are conducted on a dedicated Linux server with the following configuration:

- **CPU**: 13th-Gen Intel® Core™ i9-13900KF;
- **GPU**: NVIDIA® RTX 4090 (24 GB);
- **RAM**: 64 GB DDR5;
- **Software**: Python 3.11, PyTorch 2.1, CUDA/cuDNN 12.1.

D. Evaluation Metrics

We report two complementary performance metrics:

(i) **Coefficient of Determination (R^2)**. This metric quantifies the proportion of variance in the ground truth explained by the model:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (1)$$

where y_i , $\hat{y}_i$, and $\bar{y}$ denote the actual value, predicted value, and mean of the observations, respectively.

TABLE IV
KEY MATHEMATICAL SYMBOLS IN SECTION III

Symbol(s)	Dim. / Type	Description
<i>Temporal (GRU) variables</i>		
$\mathbf{x}_t$	$\mathbb{R}^{N_s}$	Ridership input at time t (one value per station).
$\mathbf{h}_{t-1}, \mathbf{h}_t$	$\mathbb{R}^{d_h}$	Previous and current hidden states.
$\mathbf{z}_t, \mathbf{r}_t, \tilde{\mathbf{h}}_t$	$\mathbb{R}^{d_h}$	Update gate, reset gate, and candidate activation.
$\{\mathbf{W}, \mathbf{U}, \mathbf{b}\}_{(\cdot)}$	matrices / vectors	Trainable weights and biases for GRU gates.
<i>Spatial (GCN / GAT) variables</i>		
$\mathbf{H}^{(l)}$	$\mathbb{R}^{N_s \times d_l}$	Node-feature matrix at layer l .
$\mathbf{W}^{(l)}$	$\mathbb{R}^{d_l \times d_{l+1}}$	GCN weight matrix at layer l .
$\mathbf{A}, \tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$	$\{0, 1\}^{N_s \times N_s}$	Adjacency matrix without / with self-loops.
$\tilde{\mathbf{D}}$	diag. matrix	Degree matrix of $\tilde{\mathbf{A}}$.
$e_{ij}^{(l)}, \mathbf{W}_{Eij}^{(l)}$	scalar	Raw / normalized GAT attention weight for edge (i, j) .
$\mathbf{a}^{(l)}$	vector	Learnable attention vector.
$\mathcal{N}(i)$	set	Neighbourhood of node i .
<i>Decomposition & dynamic weights</i>		
$\mathbf{X}_o$	$\mathbb{R}^{N_s \times N_i}$	Recent ridership input window (length N_i).
$\mathbf{X}_t, \mathbf{X}_r$	same	Trend and residual components of $\mathbf{X}_o$.
$\mathbf{X}_h$	$\mathbb{R}^{N_s \times N_h}$	Historical ridership aligned to prediction time.
$\mathbf{W}_E$	$\mathbb{R}^{N_s \times N_s}$	Dynamic edge-weight matrix from GAT.
<i>Aggregation & forecasting</i>		
$\mathbf{h}_i$	$\mathbb{R}^{d_h}$	Spatio-temporal embedding of station i .
$\mathbf{T}_{1:I}, \mathbf{P}_{I+1:I+N}$	sequences	Historical inputs and N -step forecasts.
<i>Constants</i>		
N_s	int	Number of stations (nodes).
N_i, N_h	ints	Recent / historical window lengths.
n	int	Model input length for one prediction.
t	int	Current time index.

TABLE V
KEY HYPERPARAMETERS ACROSS ALL MODELS

Hyperparameter	Value
Recent Input Window	20 (5 hours)
Historical Input Window	20 (same time, one week prior)
Prediction Horizon	15 minutes
Optimizer	Adam ($\eta = 10^{-3}$)
Batch Size	128

(ii) **Mean Arctangent Absolute Percentage Error (MAAPE) [36]**. MAAPE improves on the traditional MAPE by being more robust to small denominators and heavy-tailed errors:

$$\text{MAAPE} = \frac{1}{n} \sum_{i=1}^n \arctan \left(\left| \frac{y_i - \hat{y}_i}{y_i} \right| \right).$$

While R^2 measures overall model fit, MAAPE emphasizes pointwise accuracy and robustness to outliers. This dual-metric evaluation provides a comprehensive and reliable assessment of predictive performance under various operational regimes.

E. Results and Analysis

This section presents a comprehensive analysis of the performance of the proposed DST-TransitNet models compared to the benchmark models discussed earlier. We evaluate performance across different operational periods and under short-term and long-term forecasting settings.

1) *Performance Across All Stations and Scenarios*: Table VI summarizes the MAAPE loss and R^2 score (mean

TABLE VI

MAAPE LOSS AND R^2 SCORE OF MODELS ACROSS DIFFERENT OPERATIONAL PERIODS (MEAN $\pm$ SD)

Model	Normal		COVID-19		Protest	
	MAAPE	R^2	MAAPE	R^2	MAAPE	R^2
FFNN	0.121 $\pm$ 0.03	0.907 $\pm$ 0.07	0.179 $\pm$ 0.09	0.769 $\pm$ 0.29	0.173 $\pm$ 0.08	0.687 $\pm$ 0.76
LSTM	0.112 $\pm$ 0.03	0.922 $\pm$ 0.07	0.173 $\pm$ 0.09	0.809 $\pm$ 0.14	0.169 $\pm$ 0.08	0.783 $\pm$ 0.72
DLinear	0.141 $\pm$ 0.04	0.889 $\pm$ 0.07	0.195 $\pm$ 0.09	0.839 $\pm$ 0.23	0.185 $\pm$ 0.08	0.880 $\pm$ 0.07
iTransformer	0.118 $\pm$ 0.03	0.910 $\pm$ 0.08	0.184 $\pm$ 0.09	0.823 $\pm$ 0.10	0.170 $\pm$ 0.08	0.887 $\pm$ 0.07
GRUGCN	0.115 $\pm$ 0.03	0.911 $\pm$ 0.06	0.166 $\pm$ 0.09	0.852 $\pm$ 0.09	0.162 $\pm$ 0.08	0.896 $\pm$ 0.06
GraphWaveNet	0.097 $\pm$ 0.03	0.947 $\pm$ 0.05	0.156 $\pm$ 0.09	0.884 $\pm$ 0.09	0.151 $\pm$ 0.08	0.923 $\pm$ 0.05
STGNN	0.119 $\pm$ 0.04	0.926 $\pm$ 0.06	0.175 $\pm$ 0.09	0.833 $\pm$ 0.45	0.172 $\pm$ 0.09	0.858 $\pm$ 0.26
DS2TN v.1	0.094$\pm$0.03	0.956$\pm$0.04	0.152 $\pm$ 0.09	0.898 $\pm$ 0.10	0.152 $\pm$ 0.08	0.930 $\pm$ 0.05
DS2TN v.2	0.096$\pm$0.03	0.952$\pm$0.05	0.149$\pm$0.10	0.905$\pm$0.08	0.149$\pm$0.08	0.925$\pm$0.07
DS2TN (-TD)	0.100 $\pm$ 0.03	0.945 $\pm$ 0.05	0.151 $\pm$ 0.10	0.905$\pm$0.09	0.152 $\pm$ 0.08	0.933$\pm$0.05
DS2TN (+RDE)	0.096 $\pm$ 0.03	0.951 $\pm$ 0.05	0.150 $\pm$ 0.10	0.894 $\pm$ 0.10	0.149 $\pm$ 0.08	0.931 $\pm$ 0.04

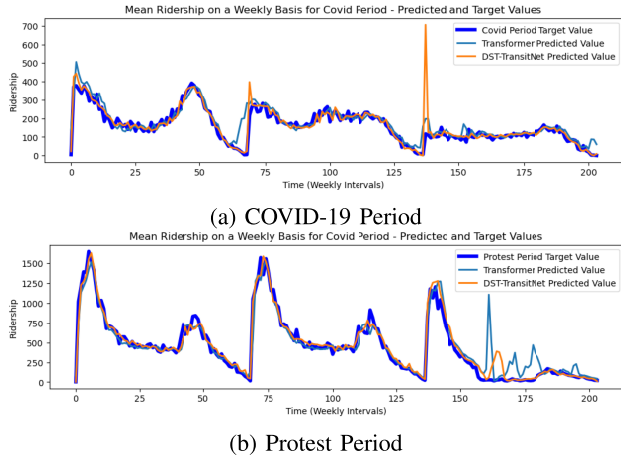


Fig. 7. Predictions by Transformer and DST-TransitNet for Station (id: 2000) during three consecutive weeks in the COVID-19 and Protest periods over a 15-minute interval.

and standard deviation) for each model across the normal, COVID-19, and protest periods.

Overall, the DST-TransitNet variants consistently achieve superior performance across all periods, confirming their robustness and predictive effectiveness. These results underscore the model's ability to capture spatio-temporal patterns and generalize well under both stable and volatile conditions.

A. Performance during disruptive events: During the COVID-19 pandemic and national protests, all models exhibit reduced performance due to increased volatility and irregular ridership patterns. As shown in Figure 7, the prediction results from the Transformer model and DS2TN v.1 model on station 2000 are compared with the target values over three selected consecutive weeks during the Covid period and Protest period. During both periods, both models perform well in the middle of the day. However, DS2TN v.1 demonstrates superior performance during peak hours, showing less deviation from the target values. Furthermore, when the system encounters unforeseen disruptions, such as the transit workers' protest starting in week 3 of the Protest period, DS2TN v.1 adapts more quickly and smoothly to the abnormal situation, providing more accurate predictions. Notably, at the start of week 3 during the COVID period, both models exhibit significant prediction errors. The models expect ridership to shift from low to high levels during the morning peak. However, due to

TABLE VII

MODEL SIZES OF SPATIO-TEMPORAL ARCHITECTURES

Model	Size
GRUGCN	479.9 KB
STGNN	481.1 KB
GraphWaveNet	1.8 MB
DS2TN v.1	1.7 MB
DS2TN v.2	482.9 KB

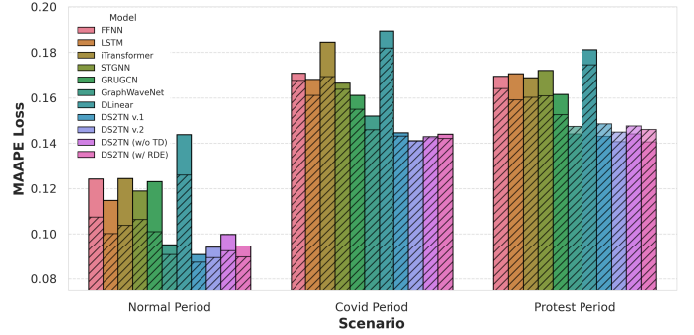


Fig. 8. MAAPE loss across models during peak and non-peak hours under normal, COVID-19, and protest periods.

COVID-19, there was no morning peak in week 3, leading to relatively large prediction errors.

Among all variants, DST-TransitNet v.2 achieves the most consistent performance, demonstrating both efficiency and adaptability—critical for real-world deployments under uncertain conditions.

B. On Recent vs. historical data for dynamic graph attention: Comparing DS2TN v.1 with DS2TN (w/ RDE), we observe that computing dynamic edge weights based on recent data yields more reliable predictions in fast-changing scenarios. This supports that recent observations provide more contextually relevant information than historical patterns in volatile environments.

C. On the impact of time decomposition: Interestingly, DS2TN (w/o TD) slightly outperforms the base model during the protest period. This is likely due to the highly irregular and fluctuating ridership patterns (e.g., unexpected zeros) introduced by the protests. In such conditions, decomposed temporal features may differ substantially from those seen during training, thereby diminishing model generalizability.

D. Computational cost: Table VII reports the model sizes of all spatio-temporal architectures considered in this study. Notably, DS2TN v.2, with its optimized structure, is approximately one-third the size of DS2TN v.1 while maintaining comparable predictive performance. This reduction in model size is especially important for practical deployment by transit agencies, particularly in on-board systems or devices with limited computational resources.

E. Summary: The results highlight the strengths of DST-TransitNet's dynamic spatio-temporal architecture in preserving performance under diverse conditions. The relatively low standard deviations further suggest that the models offer stable predictions across all stations in both stable and turbulent regimes.

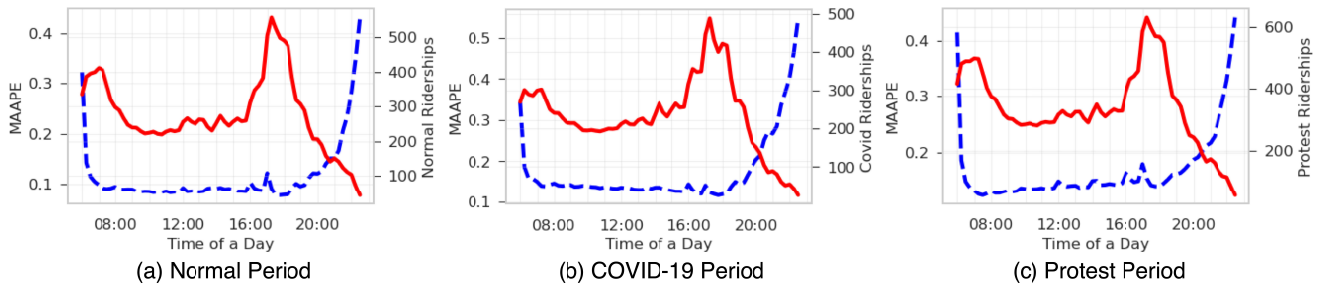


Fig. 9. MAAPE (dashed blue) and actual ridership (solid red) over 15-minute intervals from 6:00 AM to midnight across periods.

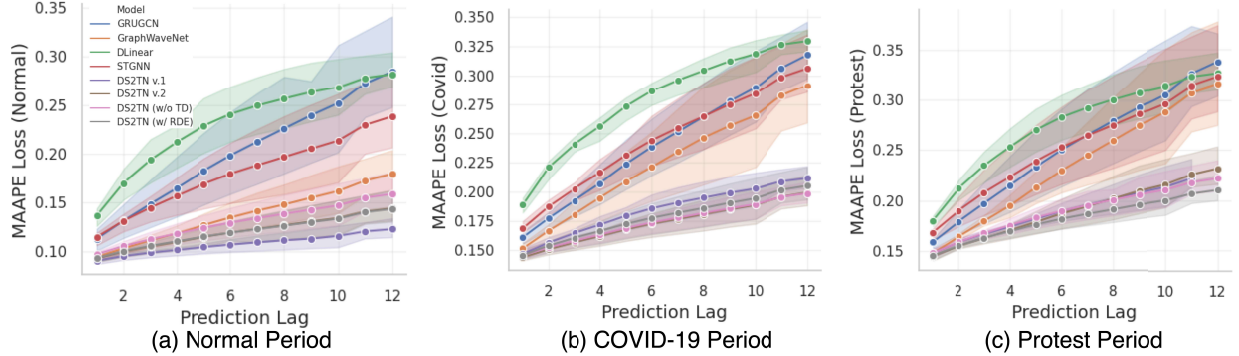


Fig. 10. MAAPE Loss Across Different Models and Prediction Lags During Normal, COVID-19, and Protest Periods. The solid line in each plot represents the loss value of each model at the corresponding prediction lag, and the shaded area indicates the range between peak and non-peak hour MAAPE values.

2) *Performance During Peak Versus Non-Peak Hours:* Figure 8 compares MAAPE loss across peak and non-peak hours. Peak hours are defined as 6:00–10:00 AM and 5:00–9:00 PM. As expected, all models perform worse during peak hours due to higher variability caused by congestion and sudden surges in ridership. Nevertheless, DST-TransitNet variants consistently outperform the baselines in both regimes.

We also observe that removing the time decomposition module results in an accuracy drop during the normal period. Additionally, during peak hours, using recent observations for graph attention (w/ RDE) underperforms compared to using historical data, possibly because the latter provides a more complete context around peak transitions, as it includes the data collected at the same time and a little after one day before.

Figure 9 further illustrates how MAAPE varies by time of day. Errors increase near the beginning and end of peak periods, and generally rise throughout the day—especially under volatile conditions. As shown in Figure 6, the lower ridership during the COVID-19 period leads to smoother patterns and smaller deviations from predictions. In contrast, the protest period is characterized by persistent ridership irregularities, which degrade predictive performance regardless of peak or non-peak status.

3) *Long-Term Forecasting Performance:* Figure 10 illustrates how MAAPE loss increases for each model as the prediction horizon extends from 1 to 12 time intervals. In our experimental setup, this corresponds to forecasting ridership from the next 15 minutes up to 3 hours.

As shown in the figure, all models exhibit increasing prediction error as the forecasting interval lengthens. This trend is expected, as prediction errors accumulate over time and propagate through the input sequence, degrading model accuracy. Nevertheless, our proposed DST-TransitNet models

consistently achieve the lowest MAAPE across all prediction lags, outperforming all baselines. Moreover, they exhibit the most stable performance as the prediction horizon increases.

While baseline models tend to show widening gaps in performance between peak and non-peak hours as prediction lags increase, DST-TransitNet maintains a narrow and stable margin. This demonstrates the strong temporal resilience and predictive robustness of our models, reinforcing their suitability for real-world applications that require reliable long-term forecasts under varying demand patterns.

4) *Temporal–Spatial Error Analysis:* The preceding subsection examined errors from a purely temporal viewpoint. Here, we jointly analyze temporal and spatial patterns to clarify *where* and *when* the model struggles, thereby guiding future improvements.

Spatial view We label the bottom-20 % of stations by R^2 as *challenge stations*. Figure 11 maps these stops against job-density data. Two concentrations emerge:

- 1) boundaries between distinct land-use zones, and
- 2) residential termini near route ends.

Edge stations show weak spatial correlation with neighbours, while residential stops exhibit demand patterns that differ from the system average. Figure 12 confirms this, showing that challenge stations feature a single sharp morning peak and an overnight data gap (00:00–06:00), both of which hamper prediction accuracy.

Temporal–spatial clustering To quantify heterogeneity, we cluster all stations via K-means on weekly ridership profiles (Fig. 13). Five distinct temporal archetypes arise:

- **Group 0** — sharp evening peak; CBD stations.
- **Group 1** — broad evening peak; mixed-use fringe.
- **Group 2** — mild single peak; transitional areas.

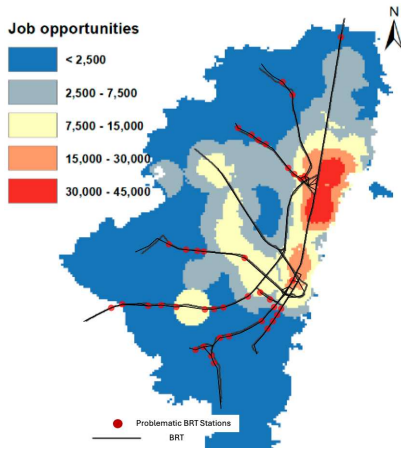


Fig. 11. Geographical distribution of challenge BRT stations in relation to job opportunities. Areas with different job opportunity levels are shown, with challenge stations highlighted in red. (Background map adapted from [37]).

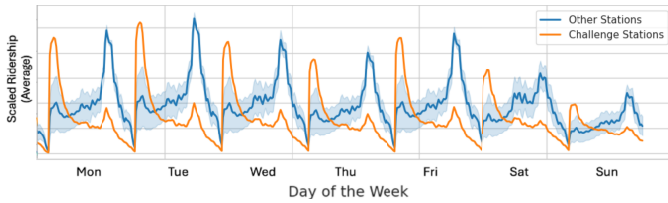


Fig. 12. Comparison of average scaled weekly ridership between challenge stations and the rest of the system.

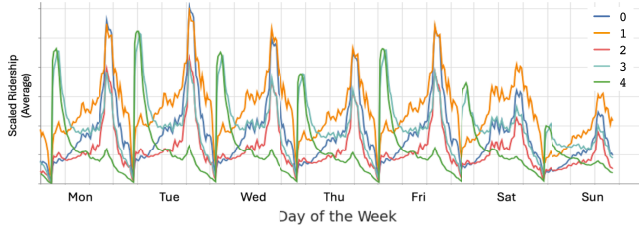


Fig. 13. Average weekly scaled ridership for station clusters derived from KMeans clustering.

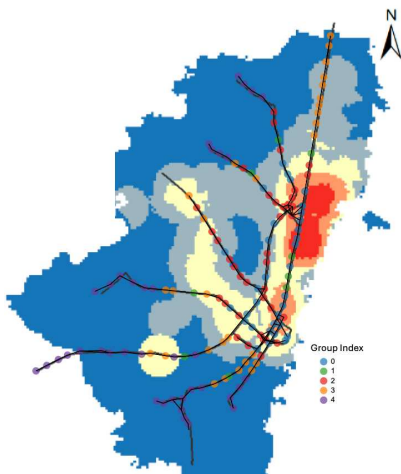


Fig. 14. Geographical distribution of station clusters overlaid on job density. (Background map adapted from [37].)

- **Group 3** — classic AM/PM peaks; system average.
- **Group 4** — single morning peak; outer residential.

Figure 14 shows these clusters align well with land-use and job-density gradients. Model performance (Fig. 15) declines

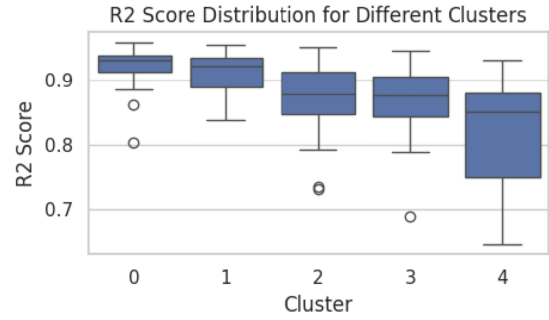


Fig. 15. R^2 score distribution across station clusters. Performance degrades from Group 0 to Group 4, with Group 4 containing most of the challenging stations.

monotonically from Group 0 to Group 4; nearly all challenge stations belong to Group 4, underscoring the difficulty of peripheral, low-density areas.

Implications Temporal and spatial effects are tightly coupled: atypical demand shapes and weak network connectivity co-occur at poorly performing stops. Two remedies follow:

- 1) **Cluster-aware training:** train or fine-tune models within homogeneous station groups to reduce feature interference.
- 2) **Context features:** incorporate land-use, job-density, and first/last-trip indicators to capture the unique dynamics of residential termini and edge zones.

These strategies should enhance robustness and narrow the accuracy gap across the network.

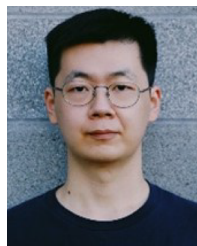
V. CONCLUSION AND FUTURE WORK

This study introduced DST-TransitNet, a hybrid spatio-temporal framework that fuses GCN/GAT, GRU, and FFNN blocks for network-wide, station-level ridership prediction. Across normal, protest, and COVID-19 regimes, the model consistently surpassed classical and state-of-the-art baselines in both short- and long-term horizons. It remained stable during peak and off-peak periods, adapted quickly to disruptive events, and sustained three-hour forecasts suitable for real-time use through a simple iterative scheme. Spatial diagnostics further revealed how accuracy varies by station cluster, guiding targeted refinements and deployment strategies.

Future work will enhance feature engineering using trend-aware temporal blocks, richer node embeddings, and exogenous signals such as weather, events, and socioeconomic indicators. We will also explore novel model architectures to further improve performance. This work was completed in early 2024, using models that were state-of-the-art at the time. Given the rapid development in spatio-temporal forecasting, future evaluations will benchmark DST-TransitNet against more recent models. To improve robustness, we will explore anomaly-aware and meta-learning techniques, and examine the model's transferability to other time-series domains (e.g., energy, traffic). Integration into transit management platforms remains a priority, enabling real-time monitoring, anomaly detection, and proactive control. We will also investigate hierarchical or multi-resolution forecasting to reduce long-horizon error accumulation, and enhance explainability through contextual data such as land use and census information.

REFERENCES

- [1] A. Boukerche and J. Wang, "Machine learning-based traffic prediction models for intelligent transportation systems," *Comput. Netw.*, vol. 181, Nov. 2020, Art. no. 107530.
- [2] P. Redhu and K. Kumar, "Short-term traffic flow prediction based on optimized deep learning neural network: PSO-Bi-LSTM," *Phys. A, Stat. Mech. Appl.*, vol. 625, Sep. 2023, Art. no. 129001.
- [3] L. Deepa, A. Rawoof Pinjari, S. Krishna Nirmale, K. K. Srinivasan, and T. Rambha, "A direct demand model for bus transit ridership in Bengaluru, India," *Transp. Res. Part A: Policy Pract.*, vol. 163, pp. 126–147, Sep. 2022.
- [4] V. Stover and E. McCormack, "The impact of weather on bus ridership in Pierce County, Washington," *J. Public Transp.*, vol. 15, no. 1, pp. 95–110, Mar. 2012.
- [5] R. Cervero, J. Murakami, and M. Miller, "Direct ridership model of bus rapid transit in Los Angeles County, California," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 2145, no. 1, pp. 1–7, Jan. 2010.
- [6] J. Gutiérrez, O. D. Cardozo, and J. C. García-Palomares, "Transit ridership forecasting at station level: An approach based on distance-decay weighted regression," *J. Transp. Geography*, vol. 19, no. 6, pp. 1081–1092, Nov. 2011.
- [7] E. Diab, D. Kasraian, E. J. Miller, and A. Shalaby, "The rise and fall of transit ridership across canada: Understanding the determinants," *Transp. Policy*, vol. 96, pp. 101–112, Sep. 2020.
- [8] C. Liu, S. Erdogan, T. Ma, and F. W. Ducca, "How to increase rail ridership in Maryland: Direct ridership models for policy guidance," *J. Urban Planning Develop.*, vol. 142, no. 4, Dec. 2016, Art. no. 04016017.
- [9] J. D. Caicedo, C. Guirado, M. C. Gonz lez, and J. L. Walker, "Share, collaborate, benchmark: Advancing travel demand research through rigorous open-source collaboration," 2023, *arXiv:2306.06194*.
- [10] C.-H. Cheng, M.-C. Tsai, and Y.-C. Cheng, "An intelligent time-series model for forecasting bus passengers based on smartcard data," *Appl. Sci.*, vol. 12, no. 9, p. 4763, May 2022.
- [11] Y. Liu, Z. Liu, and R. Jia, "DeepPF: A deep learning based architecture for metro passenger flow prediction," *Transp. Res. C, Emerg. Technol.*, vol. 101, pp. 18–34, Apr. 2019.
- [12] L. Bai, L. Yao, S. S. Kanhere, Z. Yang, C. Jing, and X. Wang, "Passenger demand forecasting with multi-task convolutional recurrent neural networks," in *Proc. Pacific-Asia Conf. Knowl. Discovery Data Mining*. Cham, Switzerland: Springer, 2019, pp. 29–42.
- [13] C. Li, L. Bai, W. Liu, L. Yao, and S. T. Waller, "Graph neural network for robust public transit demand prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 5, pp. 4086–4098, May 2022.
- [14] J. Wu, X. Li, D. He, Q. Li, and W. Xiang, "Learning spatial-temporal dynamics and interactivity for short-term passenger flow prediction in urban rail transit," *Int. J. Speech Technol.*, vol. 53, no. 16, pp. 19785–19806, Aug. 2023.
- [15] X. Wang, X. Xu, Y. Wu, and J. Liu, "An effective spatiotemporal deep learning framework model for short-term passenger flow prediction," *Soft Comput.*, vol. 26, no. 12, pp. 5523–5538, Jun. 2022.
- [16] D. Da Silva and A. Shalaby, "Forecasting short-term subway passenger flow using Wi-Fi data: Comparative analysis of advanced time-series methods," *J. Intell. Transp. Syst.*, vol. 2024, pp. 1–21, Nov. 2024.
- [17] Y. Liu et al., "ITransformer: Inverted transformers are effective for time series forecasting," in *Proc. 12th Int. Conf. Learn. Represent.*, 2024.
- [18] A. Zeng, M. Chen, L. Zhang, and Q. Xu, "Are transformers effective for time series forecasting?," in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, 2023, pp. 11121–11128.
- [19] Q. Yang, X. Xu, Z. Wang, J. Yu, and X. Hu, "Are graphs and GCNs necessary for short-term metro ridership forecasting?," *Expert Syst. Appl.*, vol. 254, Nov. 2024, Art. no. 124431.
- [20] A. Ali, Y. Zhu, Q. Chen, J. Yu, and H. Cai, "Leveraging spatio-temporal patterns for predicting citywide traffic crowd flows using deep hybrid neural networks," in *Proc. IEEE 25th Int. Conf. Parallel Distrib. Syst. (ICPADS)*, Dec. 2019, pp. 125–132.
- [21] A. Ali, Y. Zhu, and M. Zakarya, "A data aggregation based approach to exploit dynamic spatio-temporal correlations for citywide crowd flows prediction in fog computing," *Multimedia Tools Appl.*, vol. 80, no. 20, pp. 31401–31433, Aug. 2021.
- [22] A. Ali, Y. Zhu, and M. Zakarya, "Exploiting dynamic spatio-temporal graph convolutional neural networks for citywide traffic flows prediction," *Neural Netw.*, vol. 145, pp. 233–247, Jan. 2022.
- [23] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, "Graph WaveNet for deep spatial-temporal graph modeling," in *Proc. 28th Int. Joint Conf. Artif. Intell.*, Aug. 2019, pp. 1907–1913, doi: 10.24963/ijcai.2019/264.
- [24] Y. Li, R. Yu, C. Shahabi, and Y. Liu, "Diffusion convolutional recurrent neural network: Data-driven traffic forecasting," in *Proc. Int. Conf. Learn. Represent.*, 2018.
- [25] Y. Ma, H. Lou, M. Yan, F. Sun, and G. Li, "Spatio-temporal fusion graph convolutional network for traffic flow forecasting," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102196.
- [26] A. Ali, Y. Zhu, and M. Zakarya, "Exploiting dynamic spatio-temporal correlations for citywide traffic flow prediction using attention based neural networks," *Inf. Sci.*, vol. 577, pp. 852–870, Oct. 2021.
- [27] M. Li and Z. Zhu, "Spatial-temporal fusion graph neural networks for traffic flow forecasting," in *Proc. 35th AAAI Conf. Artif. Intell.*, 2021, pp. 4189–4196.
- [28] A. Cini, I. Marisca, D. Zambon, and C. Alippi, "Taming local effects in graph-based spatiotemporal forecasting," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 55375–55393.
- [29] J. Gao and B. Ribeiro, "On the equivalence between temporal and static equivariant graph representations," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 7052–7076.
- [30] N. Awan et al., "Modeling dynamic spatio-temporal correlations for urban traffic flows prediction," *IEEE Access*, vol. 9, pp. 26502–26511, 2021.
- [31] J. Chung,  . G l  hre, K. Cho, and Y. Bengio, "Empirical evaluation of gated recurrent neural networks on sequence modeling," in *Proc. NIPS Workshop Deep Learn.*, Dec. 2014.
- [32] T. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Represent.*, 2017.
- [33] C. Morris et al., "Weisfeiler and Leman go neural: Higher-order graph neural networks," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, 2019, pp. 4602–4609.
- [34] P. Veli  kovi  , G. Cucurull, A. Casanova, A. Romero, P. Li  , and Y. Bengio, "Graph attention networks," in *Proc. Int. Conf. Learn. Represent.*, 2018.
- [35] P. D. Lemoine et al., "TransMilenio, a scalable bus rapid transit system for promoting physical activity," *J. Urban Health*, vol. 93, no. 2, pp. 256–270, Apr. 2016.
- [36] S. Kim and H. Kim, "A new metric of absolute percentage error for intermittent demand forecasts," *Int. J. Forecasting*, vol. 32, no. 3, pp. 669–679, Jul. 2016.
- [37] L. A. Guzman, D. Oviedo, and R. Cardona, "Accessibility changes: Analysis of the integrated public transport system of Bogot  ," *Sustainability*, vol. 10, no. 11, p. 3958, Oct. 2018.



Jiahao Wang received the B.Eng. and M.Sc. degrees in computer science. He is currently pursuing the Ph.D. degree with the Department of Civil and Mineral Engineering, Transit Analytics Laboratory (TAL), University of Toronto, under the supervision of Prof. Amer Shalaby. His research interests include optimizing transit management using advanced deep learning and reinforcement learning methodologies. He is also exploring innovative applications of large language models (LLMs) in public transit.



Amer Shalaby received the B.A.Sc. degree from Ain Shams University, Egypt, in 1988, and the M.A.Sc. and Ph.D. degrees from the University of Toronto, Toronto, ON, Canada, in 1991 and 1996, respectively. Since 2001, he has been a Professor with the University of Toronto.